1. What is JSON?

JSON (JavaScript Object Notation) is a lightweight format used to store and exchange data. It's easy for humans to read and write, and easy for machines to parse and generate.

Basic Structure of JSON

JSON is built using:

- **Key–Value pairs**
- **Curly braces {}** for objects
- **Square brackets []** for arrays

```
{
  "name": "Revathy",
  "age": 22,
  "skills": ["Python", "Machine Learning", "Data Analysis"],
  "isIntern": true
}
```

Key Features

1. Lightweight and fast
2. Language-independent
3. Easy to read and write
4. Widely used in APIs and web applications

JSON Data Types

- String → "name": "Revathy"
- Number → "age": 22
- Boolean → true / false
- Array → ["Python", "ML"]
- Object → { "key": "value" }
- Null → null

Why JSON is Important?

- Used to **send data between server and client**
- Common in **APIs (like REST APIs)**
- Used in **AI/ML projects for data storage**
- Works well with Python (json module)

2. Difference between JSON and dictionary?

Feature	JSON	Python Dictionary
Type	String (text format)	Python object
Usage	Data exchange (APIs, web)	Data storage & manipulation
Syntax quotes	Uses double quotes only	Can use single or double quotes
Boolean values	true / false	True / False
Null value	null	None
Execution	Cannot run code	Can be used directly in Python
File type	.json	No specific file type

3. What is CSV file?

CSV (Comma-Separated Values) is a simple file format used to store tabular data (like Excel sheets).

Each line = one row

Values are separated by commas

Why CSV is Important?

1. Easy to create and read
2. Used in data analysis & AI/ML datasets
3. Works with tools like Excel & Python
4. Lightweight compared to other formats

4.What is virtual environment?

A virtual environment is an isolated space where you can install Python packages separately for each project.It keeps your project's dependencies independent from other projects and the system Python**.**

Key Benefits

1. Avoid version conflicts
2. Keep projects clean and organized
3. Safe experimentation (no system damage)
4. Easy to share projects (requirements.txt)

5.What is requirements.txt?

requirements.txt is a file that lists all the Python packages (libraries) your project needs, along with their versions.